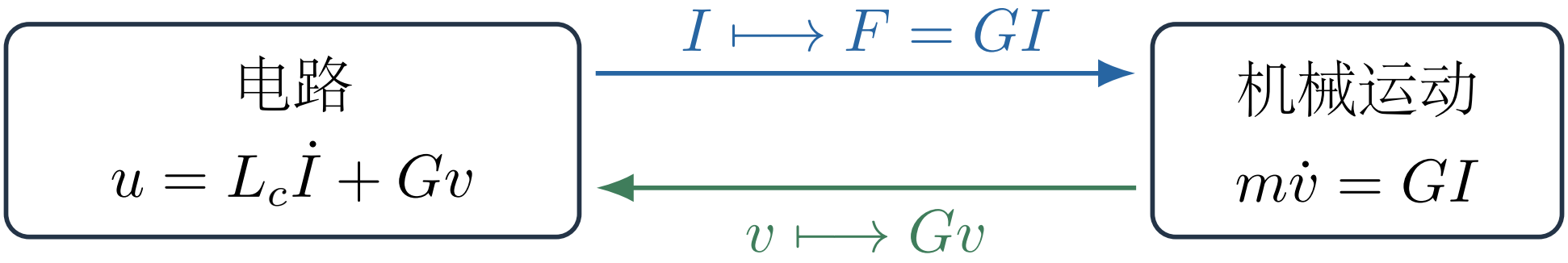

$$uI = \frac{d}{dt}\left(\frac{1}{2}mv^2 + \frac{1}{2}L_cI^2\right)$$